

FCT SERVICE NOTE

SN06080312A

Class A Service information enclosed.

Date: August 10, 2006

Subject: SCSI Tower--Remove Extra DVD Drive SCSI Termination (**JP2**) to Improve Reliability

Effectivity: SCSI Tower (DVD RAM + MOD Drives), GEMS P/N 2368838

[Applies only to GOC2, GOC3, GOC4 and GOC5 Operator Consoles]

Note: Recommended to be performed before 25365 or 25373 FMIs.

Please Note: This does NOT apply to SCSI Tower Models 2351055 or 5167410

Details: A number of SCSI Towers (of model number **2368838** only) used on Linux based systems have had double termination problem that can cause poor "SCSI" communication between Host Computer and SCSI Tower. Problems loading the INFO file, saving / restoring system state and intermittent SCSI issues can be attributed to this JP2 jumper.

Resolution: Remove the JP2 jumper on the SCSI to IDE Bridge Adapter on the DVD-RAM and check remaining jumper settings on the Adapter Board.

Attached to the rear of the DVD drive is a *SCSI to IDE Bridge Adapter Board*. The adapter board has jumper settings that must be correctly configured for proper SCSI communication. Setting the jumpers as detailed within this Service Note will improve SCSI communication with all DVD & MOD model drives.

There are two different SCSI to IDE Bridge Adapter board layouts (refer to Table 1 and Illustrations 1 and 2 for additional details).

After checking/removing the JP2 Jumper, label the SCSI Tower (on the bottom near the p/n label) with the current Month and Year as shown in the example:

Example: IDE-SCSI JP2 removed Aug 06

NOTE: Labeling the SCSI Tower (per the example) is an important step. Once the SCSI Tower cover is reinstalled, the label is the easiest method of identifying a SCSI Tower with the JP2 jumper removed.

The proper jumper setting for all versions, including 1.5, 2.0 and 3.0 of the DVD Bridge Adaptor (IDE-SCSI) board are as follows:

Table 1: SCSI Tower (2368838) Jumper Setting

Jumper	Proper SCSI Tower (2368838) Setting	Function
JP1	Set to SCSI ID # 4	See Step 4 for illustration
JP2	Removed	Termination Enable
JP3	Removed	Activity LED
JP4	Installed	Termination Power

2 Procedure

2.1 Check the SCSI Tower Device ID Numbers

1. At the Toolchest, with Application software down, open a Unix Shell and type the following:

```
{ctuser@hostname} su -
```

```
Password: #bigguy
```

```
[root@hostname] scsistat
```

NOTE: If the SCSI Tower is not recognized during the Operator Console Host Computer bootup the scsistat command will not display the SCSI Tower information. If this occurs, reboot the Operator Console.

2. Verify the output from the scsistat command resembles the following example:

Controller	SCSI ID	
Device 0	00	Direct-Access SEAGATE ST336753LW FW Rev: HPS2 (Host 36 GB operating sys drive)
Device 1	01	Direct-Access SEAGATE ST373453LW FW Rev: HPS2 (Host 73 GB Image Storage Drive#1)
Device 1	02	Direct-Access SEAGATE ST373453LW FW Rev: HPS2 (Host 73 GB Image Storage Drive#2)
Device 2	03	Optical Device SONY SMO-F551-SD FW Rev: 1.04 (Sony MOD)
Device 2	04	CD-ROM MATSHITA DVD-RAM LF-D521 FW Rev: A109 (SCSI Tower DVD-RAM)

NOTE: The DVD should be SCSI ID 4 and the Optical MOD device should be SCSI ID 3. The Controller Device maybe listed as 0 or 2 (depending on Host Computer type and software version.)

If the DVD SCSI ID is not 04, the JP1 Jumper will have to be changed.

2.2 Power-Off (Shut Down) the Console

1. Select one of the following methods to Power OFF the Operator Console:

- a. If Applications are up, click on the [Shut Down] icon and select [Shutdown].
- b. If Applications are down, open a Unix Shell at the Toolchest. Type:

```
{ctuser@hostname} halt
```

2. The Operator Console monitor will display a 'system halted' message when it is acceptable to power OFF the Operator Console.
3. Power **OFF** the Operator Console at the front panel switch.

2.3 Identify the ACARD SCSI IDE to IDE Adaptor Board Revision and Jumper Settings

1. Remove and retain the SCSI Tower cover screws.
2. Remove the SCSI Tower cover.
3. Identify the revision of the *ACARD SCSI to IDE Adapter Board* mounted to the back of the DVD drive.
4. There are three (3) revisions of the *ACARD Adapter board* used:

- rev 1.5 (refer to Illustration 1 for help identifying ACARD Adapter board revision)
- rev. 2.0 (refer to Illustration 2 for help identifying ACARD Adapter board revision)
- rev 3.0 (refer to Illustration 2 for help identifying ACARD Adapter board revision)

Illustration 1: IDE-SCSI Adapter 1.5 revision as seen attached to the back of the DVD Drive

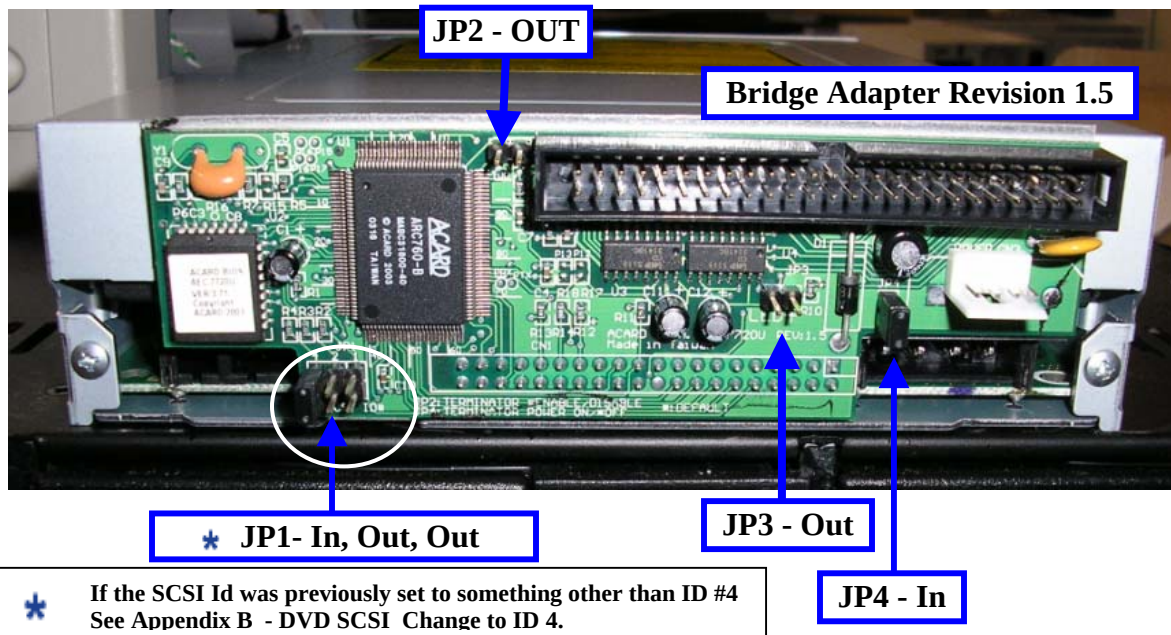


Illustration 2: IDE-SCSI Adapter 2.0 & 3.0 Revision as seen on back of the DVD Drive.



6. Check and set Jumpers as shown above and in Table 1 (detailed instructions in Appendix A).

Table 1: SCSI Tower (2368838) Jumper Setting

Jumper	Proper SCSI Tower (2368838) Setting	Function
JP1	Set to SCSI ID # 4	See Step 4 for illustration
JP2	Removed	Termination Enable
JP3	Removed	Activity LED
JP4	Installed	Termination Power

7. Remove JP2 if present. (See Appendix A for details.)

8. If JP1 is not set to ID #4, see Appendix B, 1.1 Change JP1 Jumper ID to properly change the ID #.

2.4 Reassemble the SCSI Tower

1. Reassemble SCSI Tower cover.
2. Label the SCSI Tower (on the bottom near the p/n label) with the current Month and Year:

Example: **IDE-SCSI JP2 removed Aug 06**

NOTE: Once the SCSI Tower cover is reinstalled, the label is the easiest method of identifying a SCSI Tower with the JP2 jumper removed.

2.5 Power On the Operator Console and SCSI Tower

3. Verify the SCSI Tower cables are properly connected.
4. Power **ON** the Operator Console at the front switch.
5. Verify the SCSI Tower front power LED is lit, if not power on the SCSI Tower front switch.
6. At the Toolchest, with Application software down, open a Unix Shell and type the following:

```
{ctuser@hostname} su -
```

```
Password: #bigguy
```

```
[root@hostname] scsistat
```

NOTE: If the SCSI Tower is not recognized during the Operator Console Host Computer startup the scsistat command will not display the SCSI Tower information. If this occurs, reboot the Operator Console.

7. Verify the output from the scsistat command resembles the following example:

Controller	SCSI ID	
Device 0	00	Direct-Access SEAGATE ST336753LW FW Rev: HPS2 (Host 36 GB operating sys drive)
Device 1	01	Direct-Access SEAGATE ST373453LW FW Rev: HPS2 (Host 73 GB Image Storage Drive#1)
Device 1	02	Direct-Access SEAGATE ST373453LW FW Rev: HPS2 (Host 73 GB Image Storage Drive#2)
Device 2	03	Optical Device SONY SMO-F551-SD FW Rev: 1.04 (Sony MOD)
Device 2	04	CD-ROM MATSHITA DVD-RAM LF-D521 FW Rev: A109 (SCSI Tower DVD-RAM)

NOTE: The DVD should be SCSI ID 4 and the Optical MOD device should be SCSI ID 3. The Controller Device maybe listed as 0 or 2 (depending on Host Computer type and software version.)

NOTE: If the JP1 Jumper was changed to ID # 4, please see Appendix B, 1.2 Update Directory Settings to complete the directory changes required.

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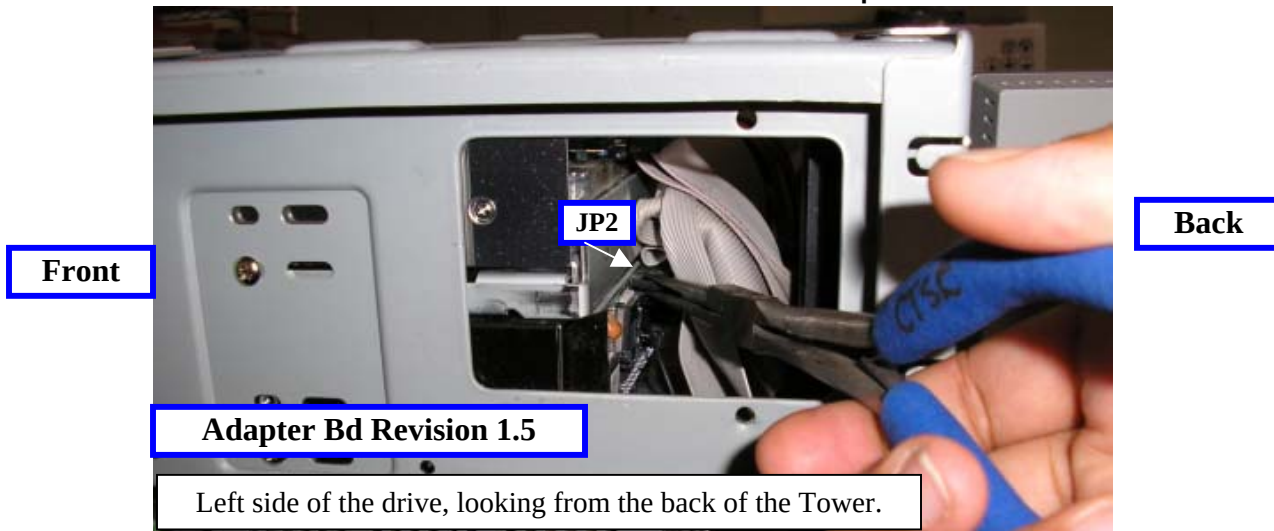
Appendix A: Removing JP2 and Setting JP1, JP3 & JP4

1.1 Verify / Remove Jumper JP2

Verify / Remove jumper JP2 per one of the following bullets depending on revision. Refer to Illustrations 3 or 4 for accessing JP2.

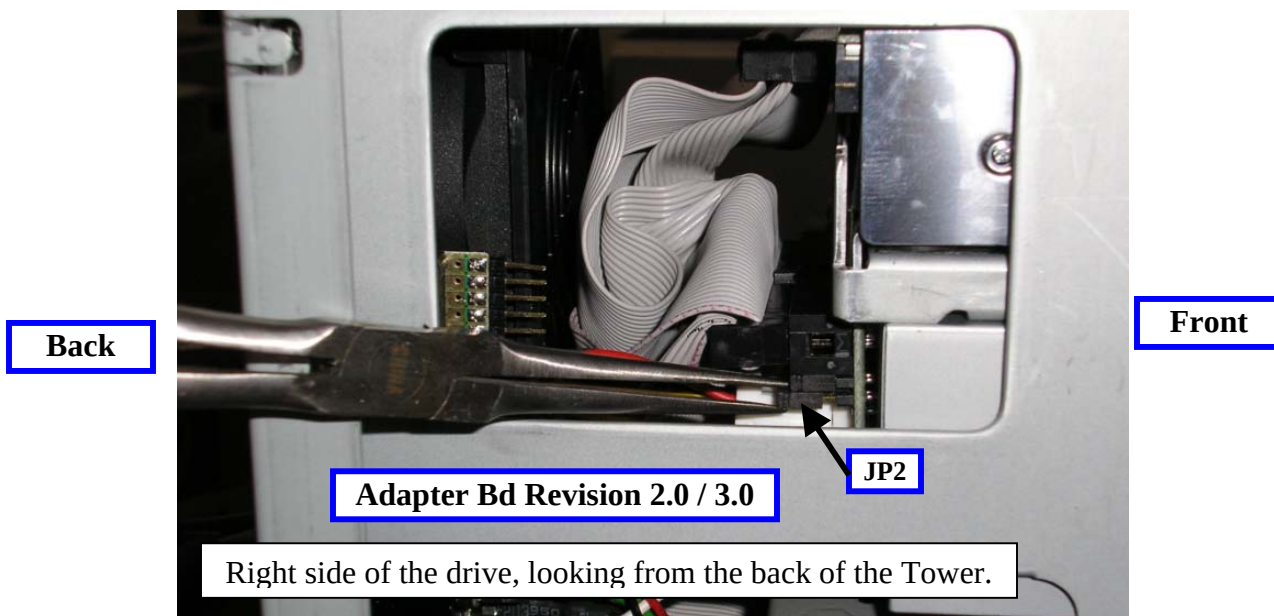
- **1.5 revision:** view the DVD Drive from rear; access JP2 from the LEFT side of the drive.

Illustration 3: IDE-SCSI Adapter 1.5 Revision



- **2.0 /3.0 revision:** view the DVD Drive from rear; access JP2 from the RIGHT side.

Illustration 4: IDE-SCSI Adapter 2.0 / 3.0 Revision



1.2 Jumpers JP1 and JP3

1. Verify / Set JP1 to SCSI ID "4". Refer to Illustration 5 for details.
2. Remove JP3. Refer to Illustration 5 for details.

NOTE: If the SCSI ID Jumper JP1 was previously something other than ID #4 then you will no longer be able to Save System State. Refer to Appendix B - DVD SCSI Change to ID 4.

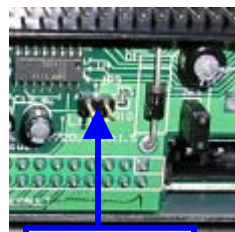
Illustration 5: JP1 SCSI ID 4

Adapter bd Rev 1.5

SCSI ID 4



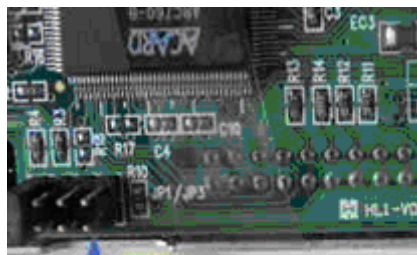
JP1- In, Out, Out



JP3 - Out

Adapter bd Rev 2.0

SCSI ID 4



JP-3 [OUT]
JP-1 [IN, OUT, OUT]

3. Verify / Install JP4. Refer to Illustration 6 or 7 for details.

Illustration 6: JP4 for 1.5 Revision

JP4 - In



Adapter Bd Rev 1.5

Illustration 7: JP4 for 2.0 / 3.0 Revision

Adapter Bd Rev 2.0 or 3.0



JP-4 [IN]
JP-2 [OUT]

Return to Section 2.3, Step 8.

Appendix B: DVD SCSI Change to ID 4

Appendix B is only needed if SCSI ID needs to be changed.

1.1 Change JP1 Jumper ID

1. Verify JP1 set to SCSI ID "4". Refer to Illustration 9 for details

NOTE: If the SCSI ID jumper JP1 was previously set to something other than ID #4, you will no longer be able to Save System State because of a software configuration in the scsi.alias file. If the SCSI ID is changed, for example from SCSI ID 0 to SCSI ID 4, the scsi.alias file still has the DVD set to SCSI ID 0 and Save System State will no longer function. A software load will correct the configuration of the scsi.alias file.

After changing the SCSI ID Jumper to ID # 4, follow the work around in section 1.2 to avoid needing a complete LFC.

Illustration 9: JP1 SCSI ID 4

Adapter bd Rev 1.5

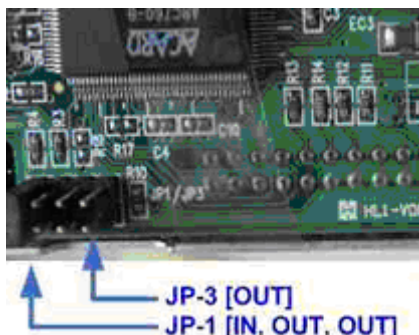
SCSI ID 4



JP1- In, Out, Out

Adapter bd Rev 2.0 or 3.0

SCSI ID 4



2. Change the JP1 Jumper to SCSI ID #4.
3. Return to section 2.4 to Re-Assemble the SCSI Tower.

1.2 Update Directory Settings

1. Change the directory to /etc as follows:

```
[root@hostname] cd /etc
```

2. Perform a more as follows:

```
[root@hostname] more scsi.alias
```

```
hostnum=2 , id=00 , devtype=cdrom, alias=dvd  
hostnum=2 , id=03 , devtype=disk, alias=mod  
hostnum=2 , id=03 , devtype=disk, partition=1, alias=mod1
```

3. Move / rename *scsi.alias* to *scsi.alias.old* as follows:

```
[root@hostname] mv scsi.alias scsi.alias.old
```

4. Recreate the *scsi.alias* file by running the *dvd_search* command as follows:

```
[root@hostname] dvd search
```

5. More out the *scsi.alias* file to view changes as follows:

```
[root@hostname] more scsi.alias
```

Example of SCSI ID 4: hostnum=2 , id=04 , devtype=cdrom, alias=dvd

6. After running the *dvd_search* command, you will be able to Save System State. A full load from cold (LFC) is recommended. The *dvd_search* command is only useful on Linux based systems.